

MBS-002 - MET Brain Structure (MBS)

A Structural Runtime Hypothesis for Embodied Cognition, Dreaming, Retrieval, Learning, and Cognitive Convergence

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Repository: <https://github.com/sizhet/MET-Brain-Structure-MBS>

Introduction

For centuries, humanity has attempted to understand the nature of the brain, cognition, learning, memory, imagination, and intelligence.

Modern neuroscience has produced an extraordinary body of knowledge, ranging from neuroanatomy and electrophysiology to brain imaging, computational neuroscience, cognitive science, and behavioral studies.

These efforts have revealed countless mechanisms, structures, and phenomena that continue to shape our understanding of biological intelligence.

This work is not intended to replace, challenge, or compete with neuroscience.

Instead, it begins with a position of respect.

The MET Brain Structure (MBS) project views modern neuroscience as the primary source of observation, evidence, and experimental knowledge regarding biological intelligence. Our goal is much narrower and more specific:

To explore whether a unified structural runtime model can help organize and interpret a wide range of cognitive phenomena through a common computational perspective.

The motivation emerged unexpectedly from a series of studies in Structural Intelligence, Function Tunnel Intelligence (FTI), Function Tunnel Math and Networks (FTMN), Calling Graph systems, Autonomous Structural Intelligence (ASI), and embodied runtime architectures.

While these projects were originally developed as engineering and

computational frameworks, they gradually revealed a recurring pattern: Across many apparently different domains, complex behavior often appeared to emerge from combinations of:

- Trigger mechanisms
- Retrieval systems
- Graph-like knowledge structures
- Functional pathways
- Runtime activation
- Continuous in-situ adaptation

The same structural pattern repeatedly appeared in software systems, autonomous agents, trajectory intelligence, tunnel-based reasoning, and eventually in observations related to biological cognition.

This led to a natural question:

Could a similar structural pattern exist within biological intelligence itself?

The MET Brain Structure hypothesis does not attempt to explain every aspect of the brain.

Instead, it focuses on a more fundamental question:

What minimal structural architecture could simultaneously support dreaming, danger response, embodied cognition, retrieval, learning, simulation, cognitive convergence, and continuous adaptation?

This question is referred to as the Minimal Evolution Threshold (MET) perspective.

Rather than searching for the most sophisticated possible architecture, the MET perspective asks:

What is the simplest architecture that evolution could have repeatedly discovered, preserved, and expanded because it provided immediate survival advantages?

From this viewpoint, many cognitive phenomena begin to appear surprisingly connected.

Dreaming is not merely a mysterious nighttime experience.

Danger response is not merely an emergency reflex.

Learning is not merely memory accumulation.

Imagination is not merely internal visualization.

Cognitive convergence among individuals is not merely cultural similarity.

Instead, these phenomena may represent different manifestations of a shared underlying structural runtime.

The central hypothesis of MBS is that biological cognition may be organized around interactions among several major components:

1. Trigger Systems

Mechanisms capable of rapidly detecting important situations, opportunities, threats, goals, or environmental changes.

2. Retrieval Structures

Systems that support fast recognition, classification, indexing, and activation of relevant knowledge structures.

3. Graph and Tunnel Knowledge Structures

Networks of states, actions, relationships, memories, embodied experiences, and functional pathways.

4. Embodied Runtime Systems

Mechanisms capable of constructing, activating, simulating, and executing meaningful scene-based experiences.

5. In-Situ Evolution Mechanisms

Continuous adaptation processes that modify triggers, retrieval pathways, graph structures, and behavioral tunnels through experience.

Under this view, cognition becomes less similar to a classical computer executing instructions and more similar to a continuously evolving structural runtime system.

Instead of:

Memory



CPU



Decision

MBS proposes a structure more closely resembling:

Trigger



Retrieval



Graph Activation



Tunnel Execution



Embodied Runtime



Learning & Evolution

One of the most important motivations for this hypothesis comes from dreaming animals.

Humans dream.

Dogs dream.

Cats dream.

Many mammals exhibit dream-like sleep behaviors.

This observation is significant because dreaming appears long before language, mathematics, formal reasoning, or human civilization.

Dreaming therefore may provide evidence for an ancient and deeply embedded runtime capability that supports embodied scene construction, memory activation, simulation, and functional pathway exploration.

If true, dreaming may not be an isolated phenomenon.

It may instead expose one of the most fundamental layers of biological intelligence.

Similarly, danger response systems may provide another critical clue.

Evolution strongly favors organisms capable of recognizing threats rapidly and activating appropriate behaviors before detailed reasoning becomes possible.

This creates intense selective pressure for efficient triggering, retrieval, classification, and pathway activation mechanisms.

From the MET perspective, such pressures may have played a major role in shaping the architecture of biological cognition.

MBS therefore proposes that many apparently independent cognitive functions may share common structural foundations.

Dreaming.

Memory recall.

Imagination.

Navigation.

Learning.

Instinct.

Danger response.

Planning.

Skill acquisition.

Cognitive convergence.

All may emerge from different forms of activation and evolution within a common trigger–retrieval–graph–tunnel runtime architecture.

The purpose of this repository is not to declare conclusions.

Its purpose is to explore possibilities.

The MBS hypothesis should be viewed as a structural framework for investigation, discussion, simulation, and future experimentation.

We hope that this perspective may serve as a bridge between neuroscience, cognitive science, artificial intelligence, embodied intelligence, and structural intelligence research.

Most importantly, we hope it encourages further collaboration with neuroscientists, cognitive researchers, psychologists, AI researchers, and

engineers whose decades of work continue to provide the essential observations upon which all future models must ultimately depend.

The goal is not replacement.

The goal is understanding.

And perhaps, through better structural understanding, a clearer path toward understanding intelligence itself.